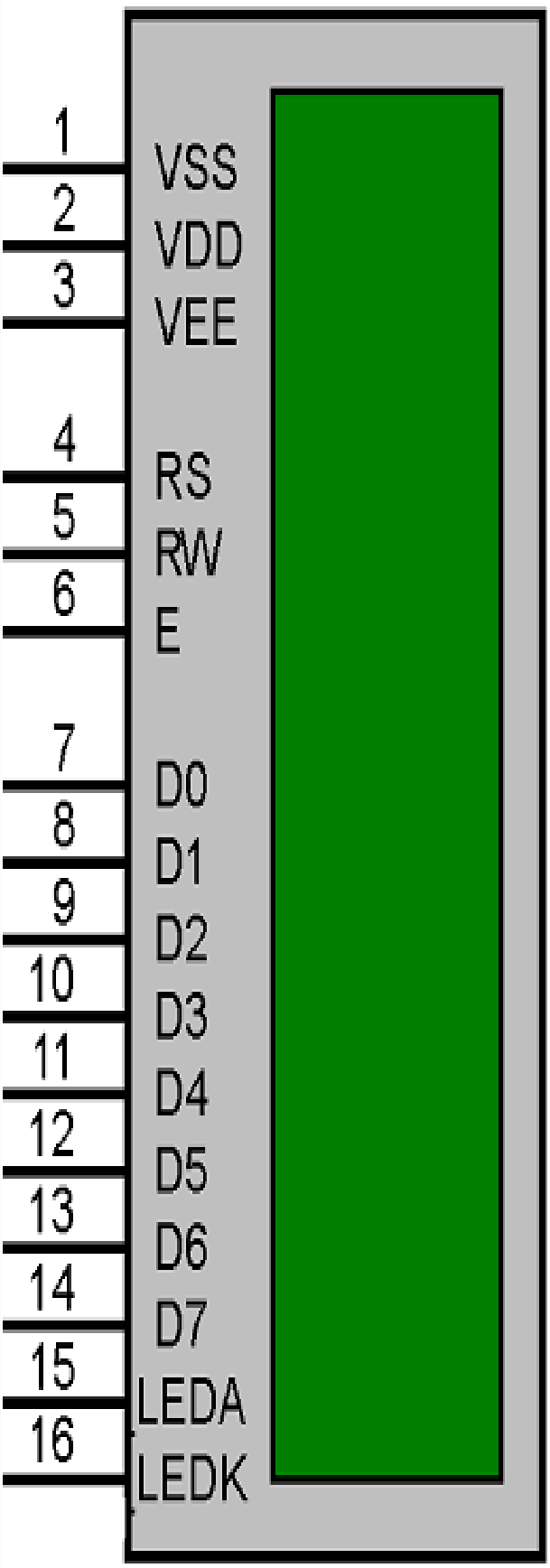
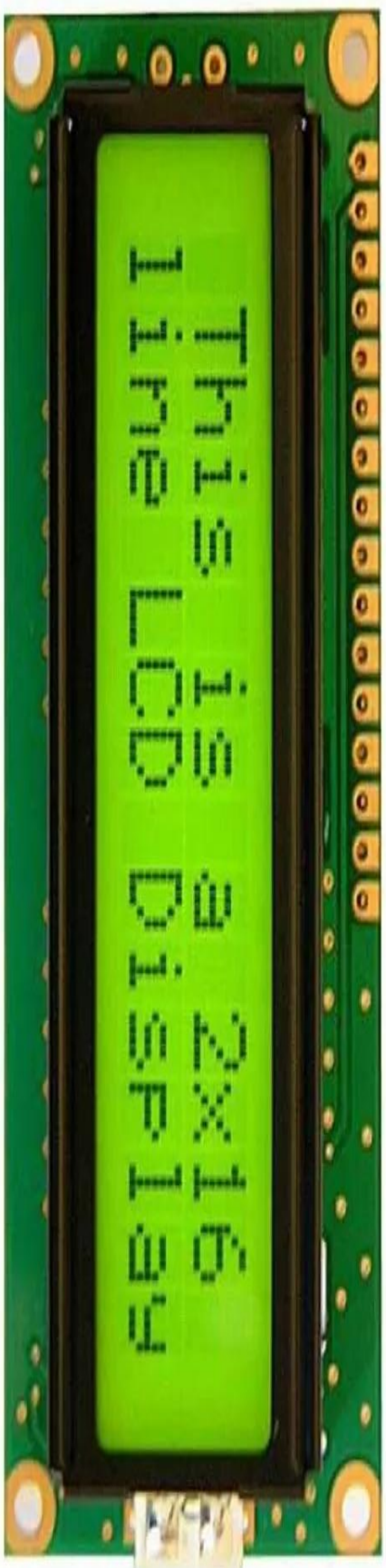
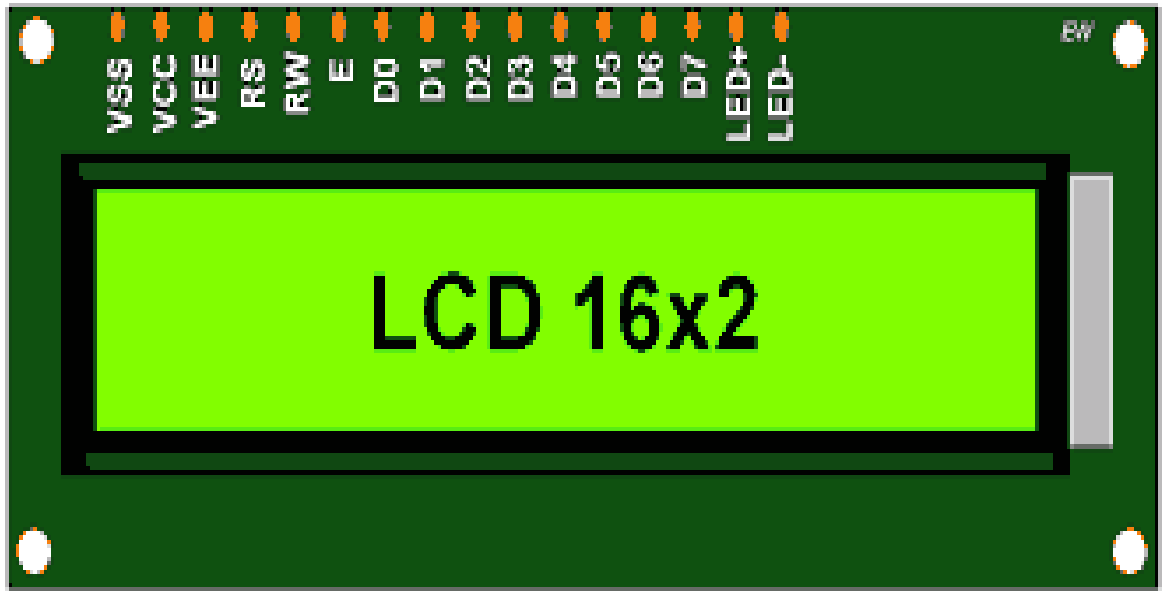


Alphanumeric/ Character LCD

Based on HD44780 Controller


1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
VSS
VDD
VEE
RS
RW
E
D0
D1
D2
D3
D4
D5
D6
D7
LEDA
LEDK

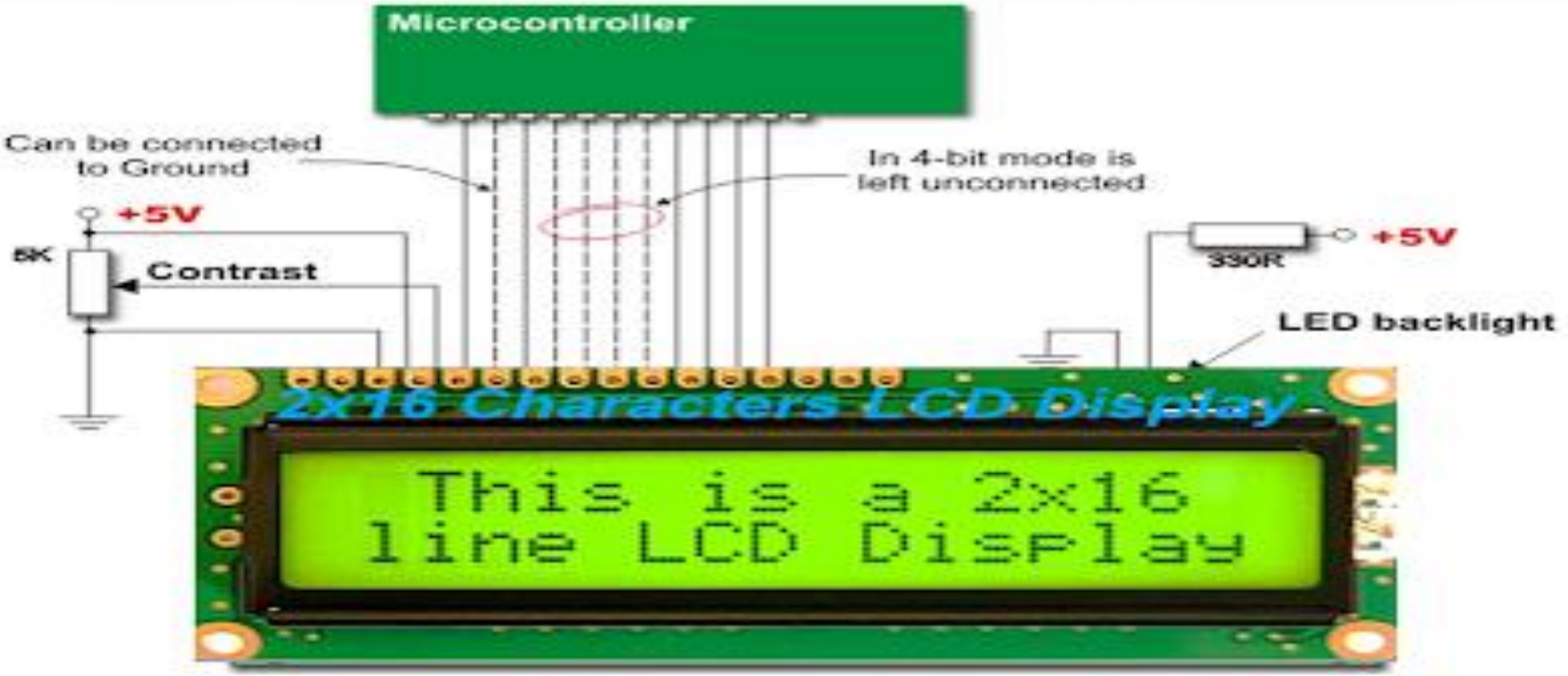




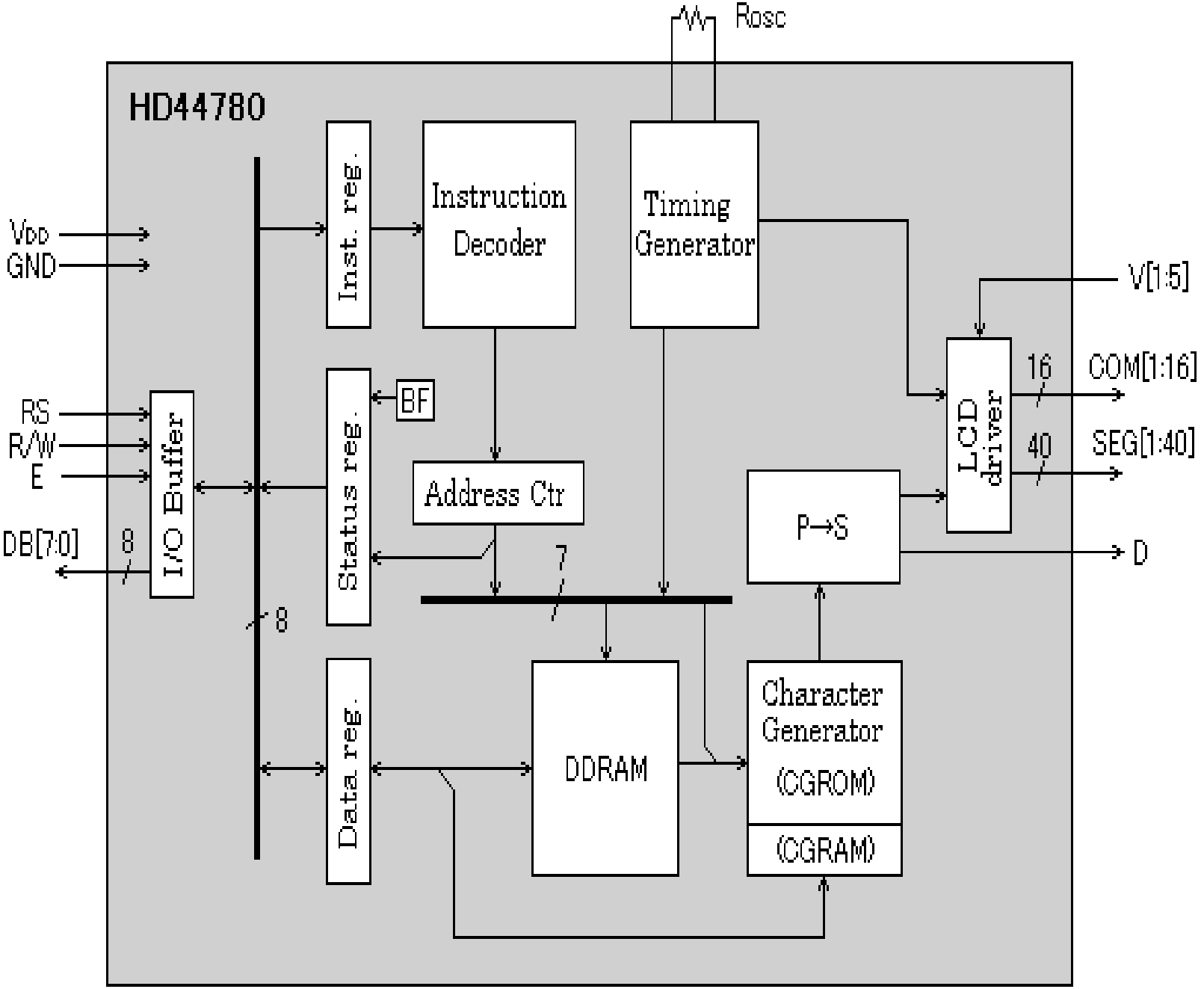
No.	PIN	Function
1	VSS	Ground
2	VCC	+5 Volt
3	VEE	Contrast control 0 Volt: High contrast.

No.	PIN	Function
4	RS	Register Select 0: Commannd Reg. 1: Data Reg.
5	RW	Read / write 0: Write 1: Read
6	E	Enable H-L pulse
7-14	D0 - D7	Data Pins D7: Busy Flag Pin
15	LED+	+5 Volt
16	LED-	Ground

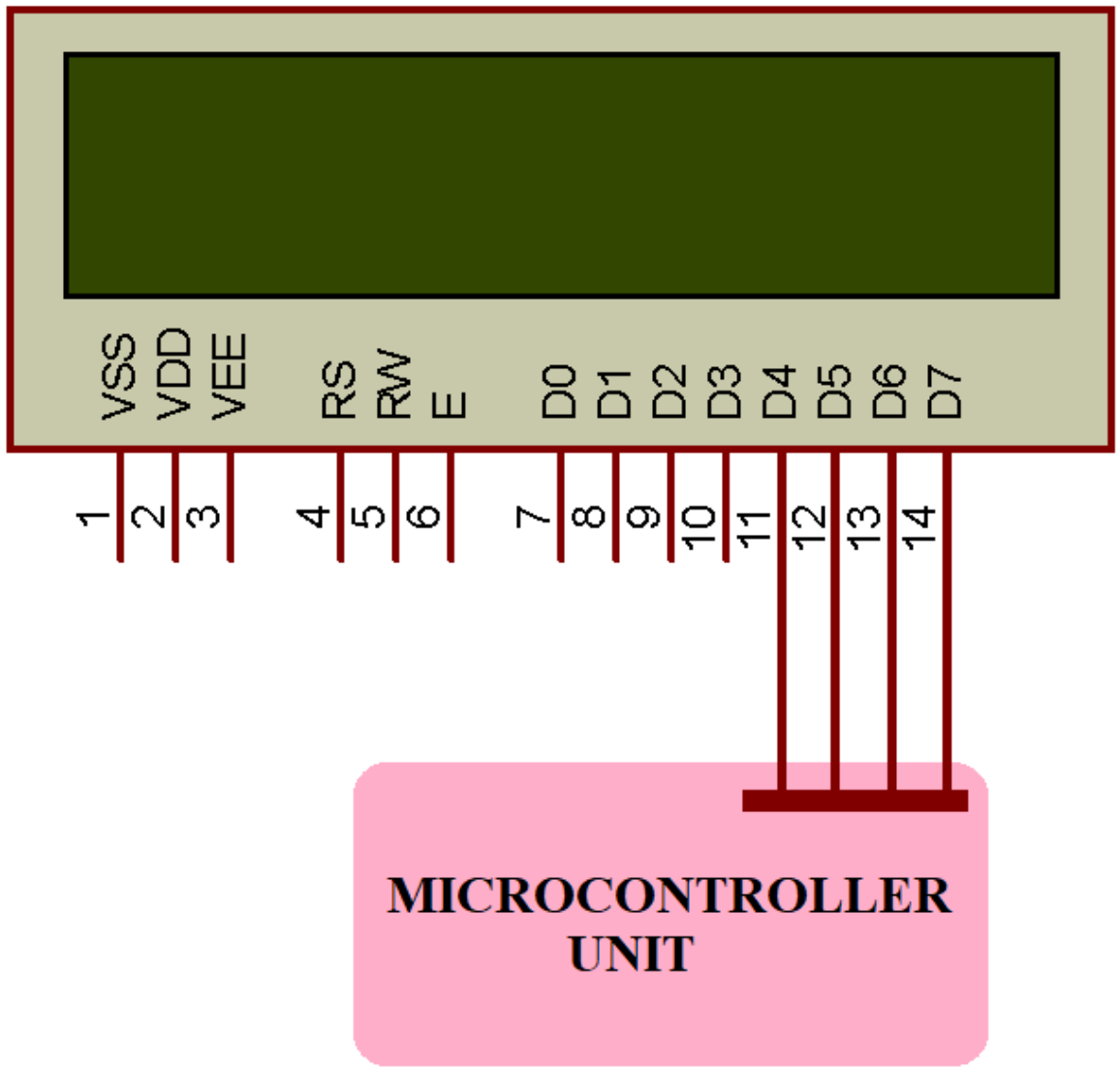

4x20 LCD DISPLAY
with HD44780
AMC2004A-B-Y6WFDY
useful for Arduino


Microcontroller
Can be connected to Ground
In 4-bit mode is left unconnected
+5V
5K
Contrast
330R
+5V
LED backlight
2x16 Characters LCD Display
This is a 2x16
line LCD Display

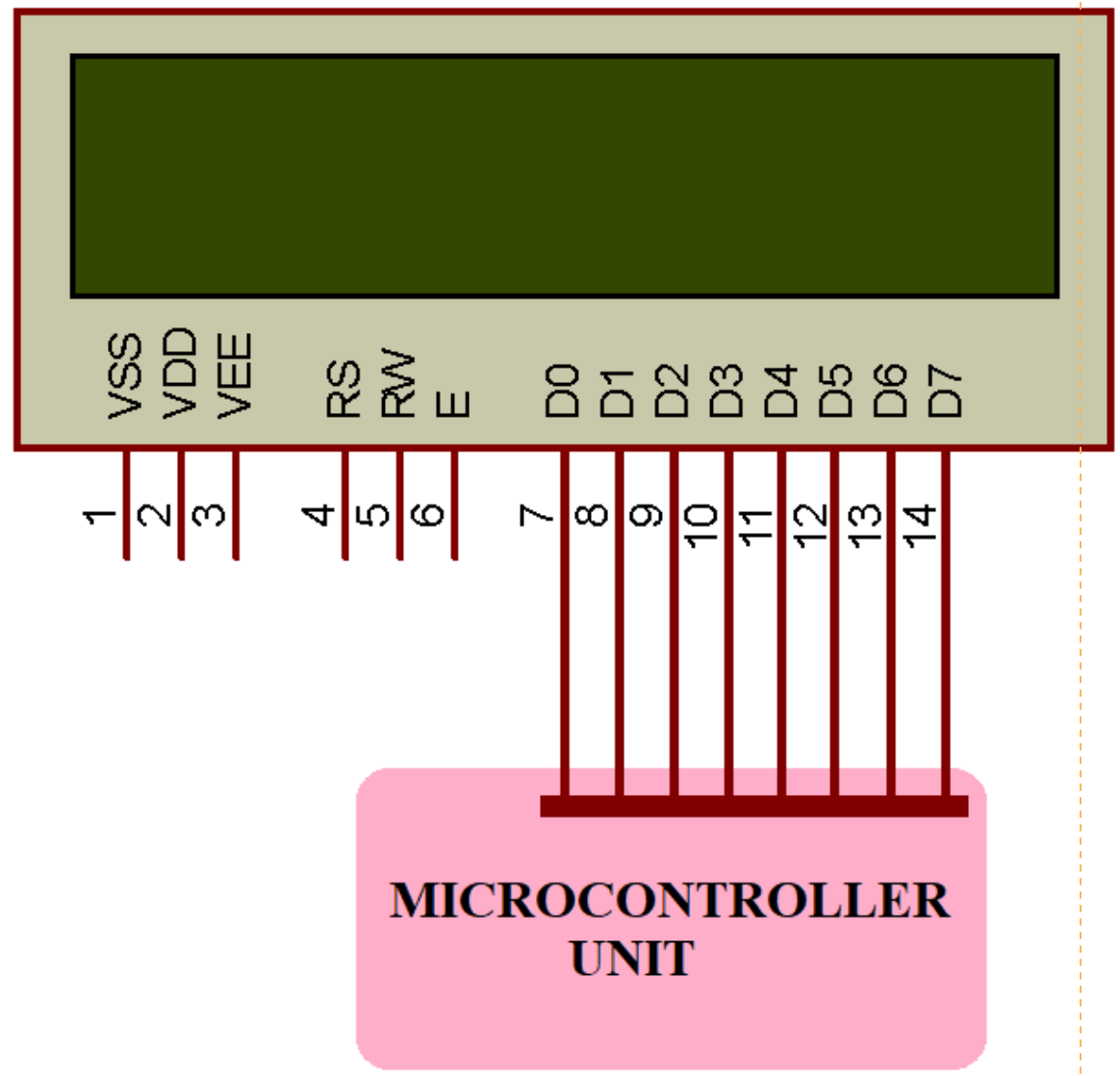

5D2
HD44780A00
JAPAN


HD44780
Rosc
Vdd
GND
RS
R/W
E
DB[7:0]
8
I/O Buffer
Inst. reg.
Instruction Decoder
Timing Generator
Status reg.
BF
Address Ctr
Data reg.
DDRAM
Character Generator
(CGROM)
(CGRAM)
P→S
LCD driver
V[1:5]
16
COM[1:16]
40
SEG[1:40]
D
7


HD44780 Controller Chip
1602A
QAPASS

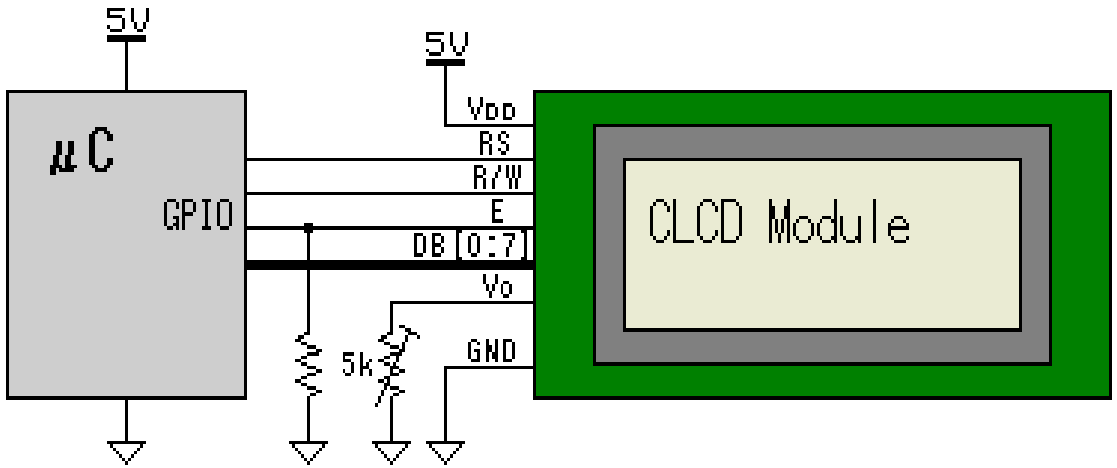


LCD : 4 BIT MODE



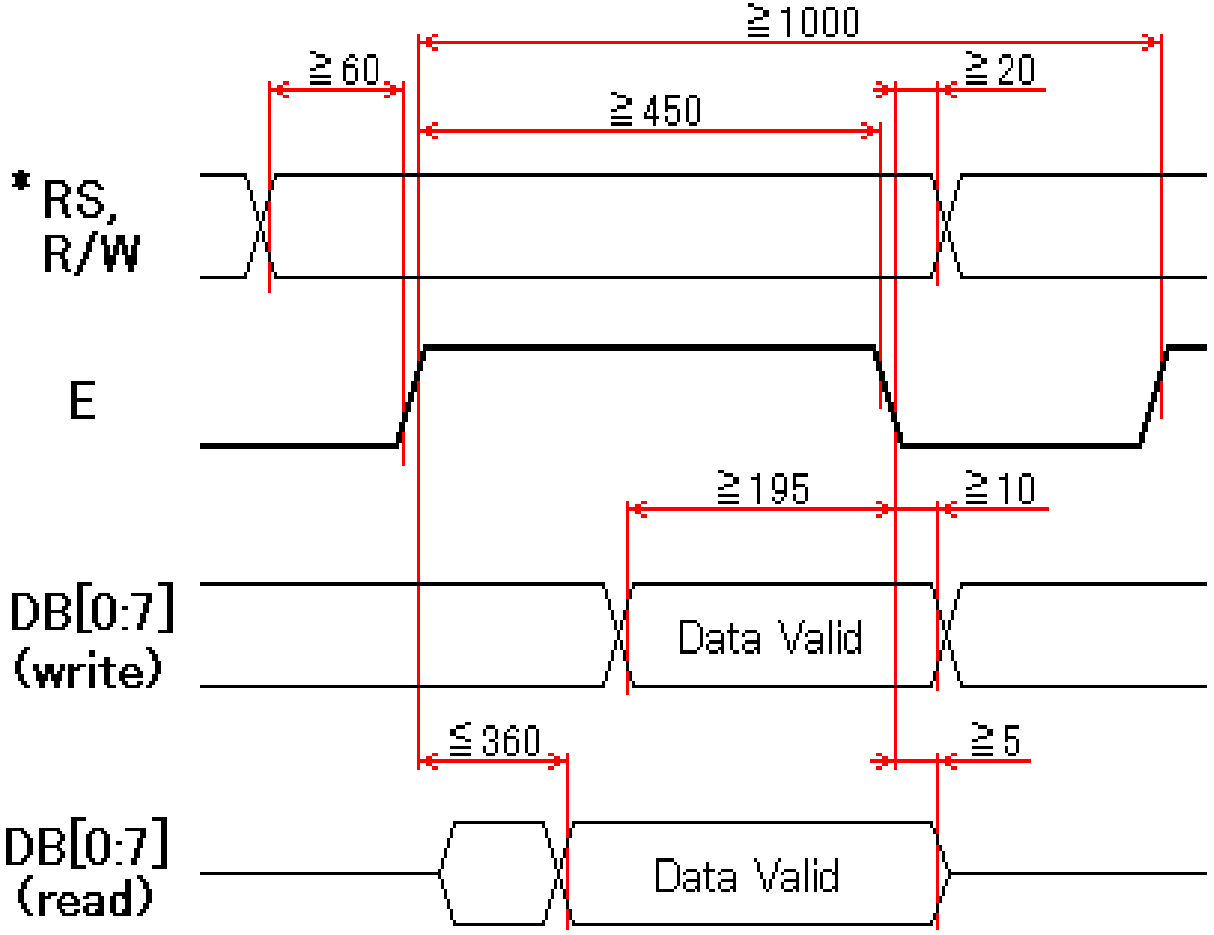
LCD : 8 BIT MODE

(a) Typical Connection between uC and LCD



Pin#	Name	Function	
1	GND	Power Supply	
2	VDD		
3	Vo	Contrast	
4	RS	Register Select	
5	R/W	Read/Write	
6	E	Enable	
7	DB0	Data Bus (8-bit mode)	N.C. (4-bit mode)
8	DB1		
9	DB2		
10	DB3		
11	DB4		Data Bus (4-bit mode)
12	DB5		
13	DB6		
14	DB7		

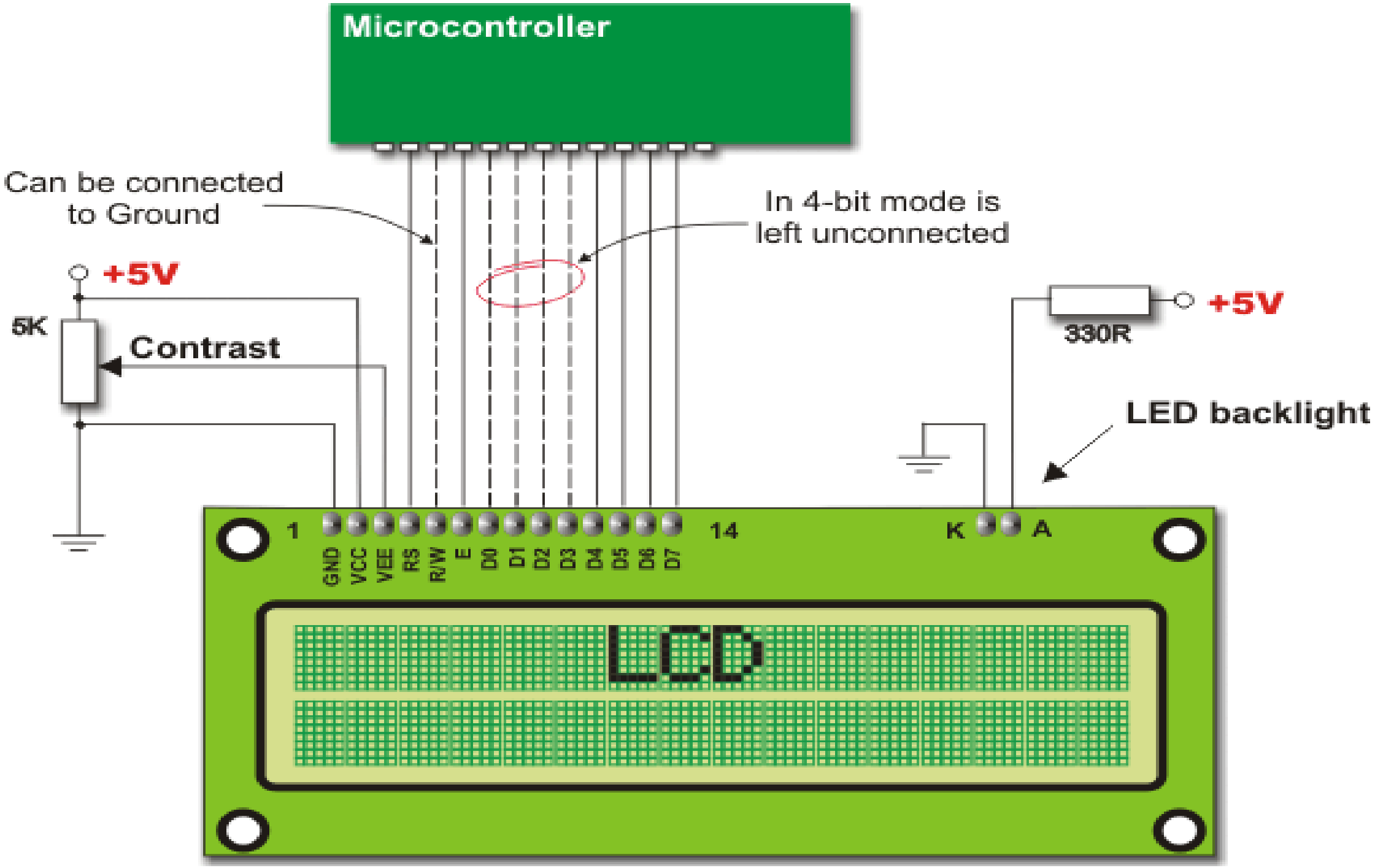
(b) Timing Diagram (1MHz)

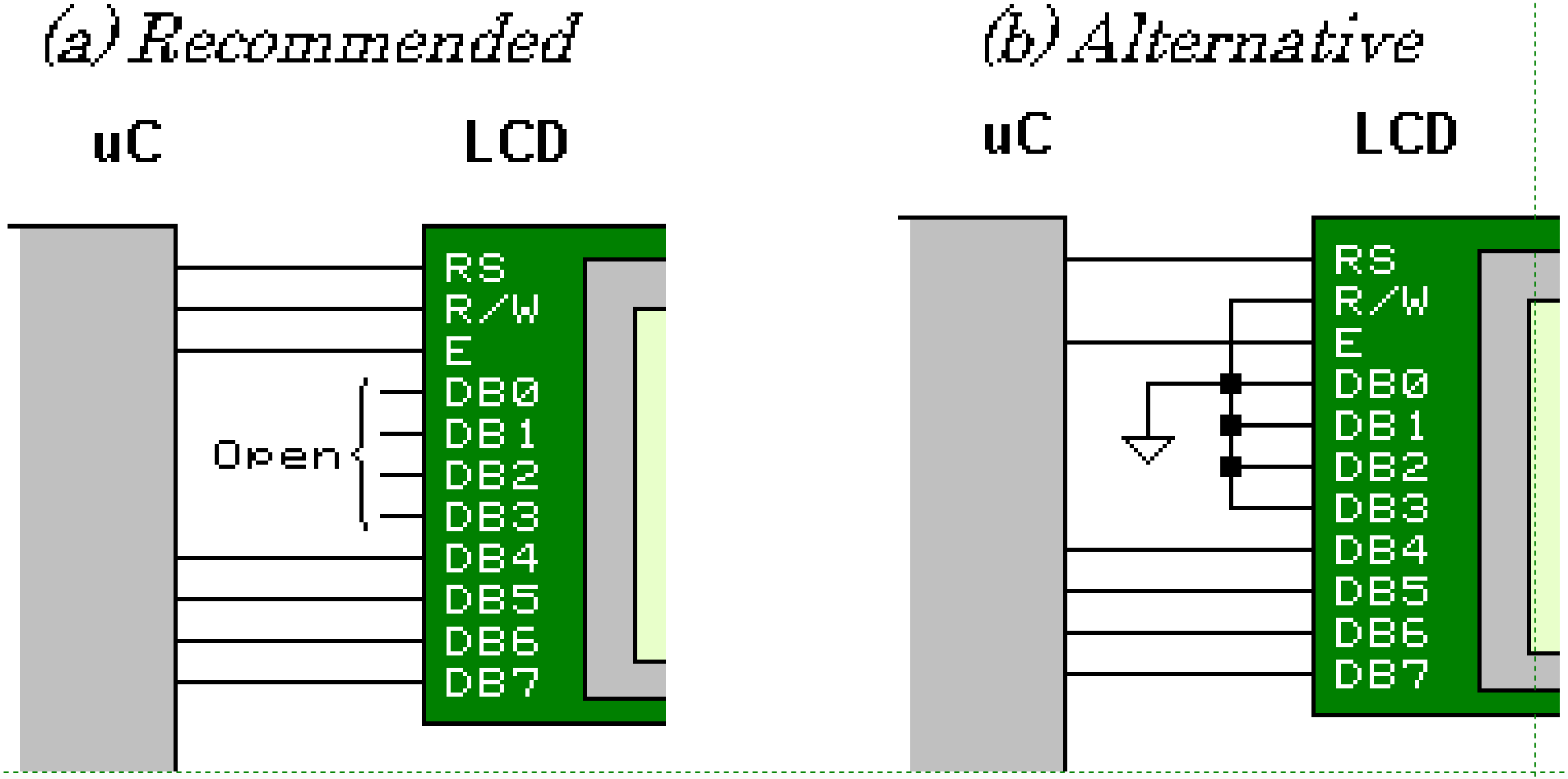


***Truth Table**

RS	R/W	Function
L	L	Write Command Register
H	L	Write Data Register
L	H	Read Status Register
H	H	Read Data Register

All units are nanosecond


Microcontroller
Can be connected
to Ground
In 4-bit mode is
left unconnected
+5V
5K
Contrast
330R
+5V
LED backlight
1
14
K
A
GND
VCC
VEE
RS
R/W
E
D0
D1
D2
D3
D4
D5
D6
D7
LCD

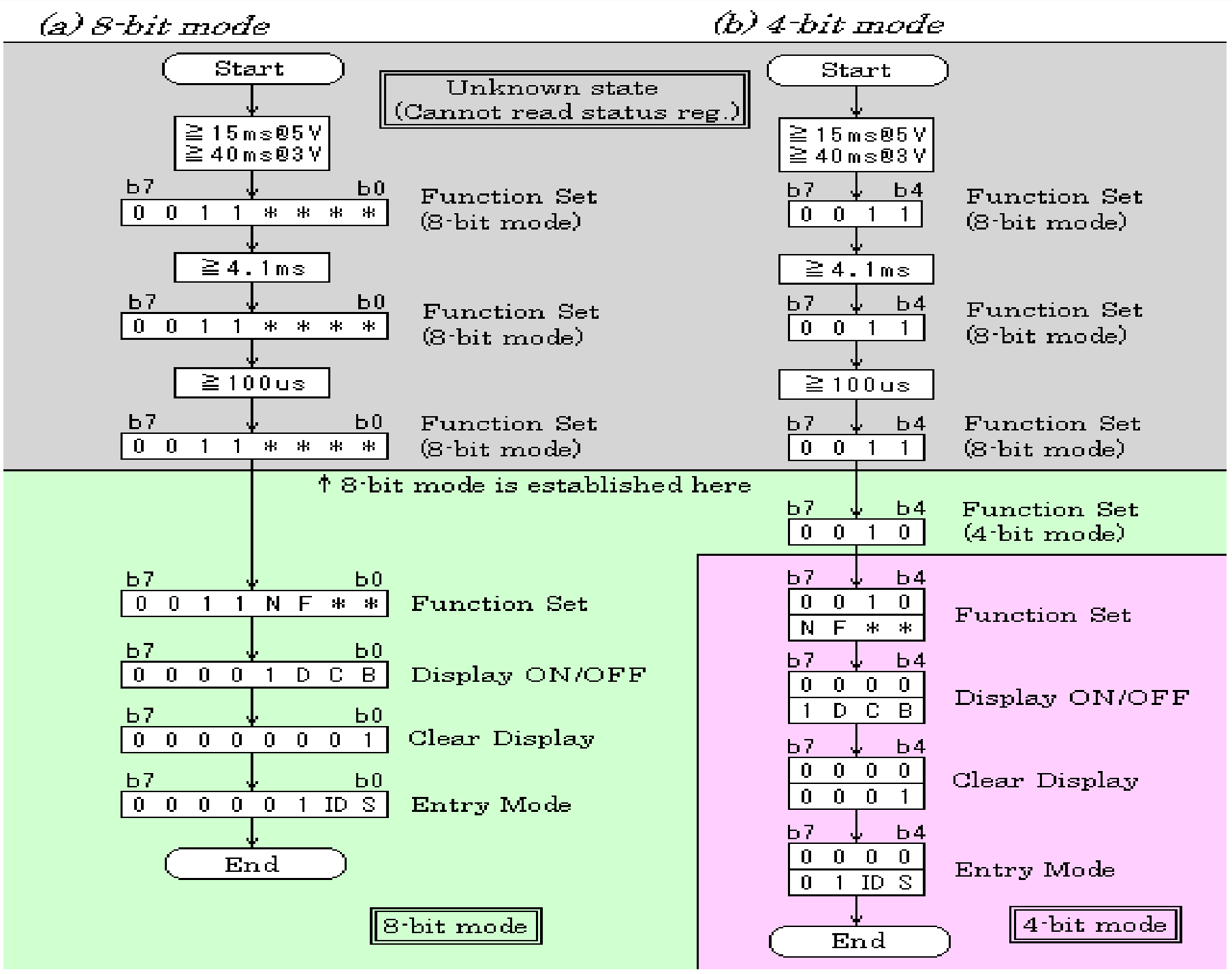

(a) Recommended
uC
LCD
RS
R/W
E
DB0
DB1
DB2
DB3
DB4
DB5
DB6
DB7
Open
(b) Alternative
uC
LCD
RS
R/W
E
DB0
DB1
DB2
DB3
DB4
DB5
DB6
DB7

Instructions

The data written into the instruction register is decoded and executed by instruction decoder. List 1 shows the each instruction.

Table 1. Instruction Code of HD44780

Instruction	Opcode								Description
	b7	b6	b5	b4	b3	b2	b1	b0	
Clear Display	0	0	0	0	0	0	0	1	Fills DDRAM with 0x20 and set DDRAM address 00h to the address counter.
Cursor Home	0	0	0	0	0	0	1	*	Sets DDRAM address 00h to the address counter. *:Don't care
Entry Mode Set	0	0	0	0	0	1	I/D	S	Sets the direction of address counter and specifies display shift (updating display offset register) on data read/write. I/D=1:Increment, S=1:With display shift
Display ON/OFF	0	0	0	0	1	D	C	B	Sets display, under-line cursor and block cursor on/off. D=1:Display ON, C=1:Under-line cursor ON, B=1:Block cursor ON
Move cursor and Shift display	0	0	0	1	S/C	R/L	*	*	Increment/decremet address counter and display offset register. S/C=1:Shift display, S/C=0:Move cursor, R/L=1:Right shift, R/L=0:Left shift
Function Set	0	0	1	DL	N	F	*	*	Configure operating mode. DL=1:8-bit bus, DL=0:4-bit bus N=1:2-row mode, N=0:1-row mode, F=1:11-line mode, F=0:8-line mode
Address Set (CGRAM)	0	1	Address(00h..3Fh)						Sets CGRAM address to the address counter. After this instruction, **CGRAM** is accessed via data register
Address Set (DDRAM)	1	Address(00h..67h)							Sets DDRAM address to the address counter. After this instruction, **DDRAM** is accessed via data register


(a) 8-bit mode
(b) 4-bit mode
Start
Unknown state
(Cannot read status reg.)
≧15ms@5V
≧40ms@3V
b7 b0
0 0 1 1 * * * *
Function Set
(8-bit mode)
≧4.1ms
≧100us
↑ 8-bit mode is established here
b7 b0
0 0 1 1 N F * *
Function Set
0 0 0 0 1 D C B
Display ON/OFF
0 0 0 0 0 0 0 1
Clear Display
0 0 0 0 0 1 ID S
Entry Mode
End
8-bit mode
b7 b4
0 0 1 1
0 0 1 0
N F * *
0 0 0 0
1 D C B
0 0 0 1
0 1 ID S
4-bit mode